

Sufficiency

Outline

- 1) Sufficiency
- 2) Factorization Theorem
- 3) Examples
- 4) Minimal sufficiency

Three models for coin flipping

Model 3 $X_{i,j} \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\theta_{i,j})$ $i=1, \dots, 48$ $j=1, \dots, n_i$ $\theta_{i,j} \rightarrow \text{in } j$

Model 2 $X_{i+} \stackrel{\text{ind}}{\sim} \text{Binom}(n_i, \theta_i)$ $X_{i+} = \sum_{j=1}^{n_i} X_{i,j}$

Model 1 $X_{++} \stackrel{\text{ind}}{\sim} \text{Binom}(n, \theta)$ $X_{++} = \sum_{i=1}^{48} \sum_{j=1}^{n_i} X_{i,j}$

most assumptions $\swarrow$ $\searrow$ fewest assumptions

These models are nested: $\mathcal{P}_1 \subseteq \mathcal{P}_2 \subseteq \mathcal{P}_3$

Data keeps getting compressed too...
are we losing anything by doing this?

Answer No. X_{++} is a sufficient statistic for $\mathcal{P}_1$, and $(X_{1+}, \dots, X_{48+})$ is also sufficient for $\mathcal{P}_2$

Def A statistic $T(X)$ is any function of data X

Sufficiency

Def A statistic $T(x)$ is sufficient for model $\mathcal{P}$ if the conditional distribution of $X \mid T(x)$ is the same for all $P \in \mathcal{P}$

Check definition for $T(x) = X_{++}$ in $\mathcal{P}_1$:

$$\begin{aligned} p_{\theta}(x) &= \prod_{i=1}^{48} \prod_{j=1}^{n_i} \theta^{x_{ij}} (1-\theta)^{1-x_{ij}} \\ &= \theta^{X_{++}} (1-\theta)^{n-X_{++}} \quad (\text{why no } \binom{n}{X_{++}}?) \end{aligned}$$

$$\begin{aligned} P_{\theta}(X=x \mid X_{++}=t) &= \frac{P_{\theta}(X=x, X_{++}=t)}{P_{\theta}(X_{++}=t)} \\ &= \frac{1\{X_{++}=t\} \cdot \theta^t (1-\theta)^{n-t}}{\binom{n}{t} \theta^t (1-\theta)^{n-t}} \\ &= 1\{X_{++}=t\} / \binom{n}{t} \end{aligned}$$

Intuition Suppose we believe Model 1.

Big/small X_{++} more likely with big/small θ

But once we know $X_{++} = 178,079$,
all data sets X with that many same-side
flips are equally likely, regardless of θ

Not true in Models 2 & 3 $\Rightarrow X_{++}$ no longer sufficient

Factorization Theorem

Usually, we can recognize sufficient stats by inspecting the density

Theorem (Factorization Theorem)

Let $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ be a model with densities $p_\theta(x)$ wrt common measure μ .

$T(x)$ is sufficient iff there exist $g_\theta(t)$, $h(x) \geq 0$ with

$$p_\theta(x) = g_\theta(T(x)) h(x) \quad (\text{for } \mu\text{-a.e. } x)$$

Note we could absorb h into μ as density

(define new base measure ν , $\nu(A) = \int_A h(x) d\mu(x)$)

$\Rightarrow \mathcal{P}$ has densities $p_\theta(x) = g_\theta(T(x))$ wrt ν

Interp: after changing base measure,

density depends on x only through $T(x)$

(Can't absorb $g_\theta(T(x))$ into μ : depends on θ)

Proof (discrete $\mathcal{X}$): Assume wlog $\mu = \#$ on $\mathcal{X}$

$$\begin{aligned} (\Leftarrow) \mathbb{P}_\theta(X=x | T=t) &= \frac{\mathbb{P}_\theta(X=x, T(x)=t)}{\mathbb{P}_\theta(T(x)=t)} \\ &= \frac{\cancel{g_\theta(t)} h(x) \mathbb{1}\{T(x)=t\}}{\sum_{T(z)=t} \cancel{g_\theta(t)} h(z)} \end{aligned}$$

$(\Rightarrow)$ Assume $T(x)$ sufficient, let

$$g_\theta(t) = \mathbb{P}_\theta(T(X)=t)$$

$$h(x) = \mathbb{P}(X=x | T(X)=T(x))$$

$\nwarrow$ no dep. on θ

$$\begin{aligned} \Rightarrow g_\theta(T(x)) h(x) &= \mathbb{P}_\theta(T(X)=T(x) \text{ and } X=x) \\ &= \mathbb{P}_\theta(X=x) = p_\theta(x) \quad \square \end{aligned}$$

Proof similar for general densities

- careful about conditioning in cts spaces

Examples

Ex. Normal location family

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\theta, 1) = \frac{1}{\sqrt{2\pi}} e^{-(x-\theta)^2/2}$$

$$\rho_\theta(x) = (2\pi)^{-n/2} \prod_{i=1}^n e^{-(x_i-\theta)^2/2}$$

$$= e^{\theta \sum x_i - n\theta^2/2} \cdot \frac{e^{-\sum x_i^2/2}}{(2\pi)^{n/2}}$$

(collect factors with
no dep. on θ)

$\Rightarrow \sum X_i$ is sufficient

Ex. Poisson family

$$X_1, \dots, X_n \stackrel{iid}{\sim} \text{Pois}(\theta) = \frac{\theta^x e^{-\theta}}{x!} \quad \text{for } x = 0, 1, 2, \dots$$

$$\rho_\theta(x) = \prod_{i=1}^n \frac{\theta^{x_i} e^{-\theta}}{x_i!} = \theta^{\sum x_i} e^{-n\theta} \cdot \frac{1}{\prod x_i!}$$

$\Rightarrow \sum X_i$ is sufficient

These two examples have something important in common!
(next lecture)

Ex. Uniform location family

$$X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{U}[\theta, \theta+1] = 1\{\theta \leq x \leq \theta+1\}$$

$$\rho_\theta(x) = \prod_{i=1}^n 1\{\theta \leq x_i \leq \theta+1\}$$

$$= 1\{\theta \leq X_{(1)}\} 1\{X_{(n)} \leq \theta+1\}$$

$\Rightarrow (X_{(1)}, X_{(n)})$ is sufficient.

Interpretations of Sufficiency

X is informative about θ only because its distribution depends on θ .

We can think of the data as being generated in two stages:

- 1) Generate T : distribution dep. on θ
- 2) Generate $X|T$: does not dep on θ

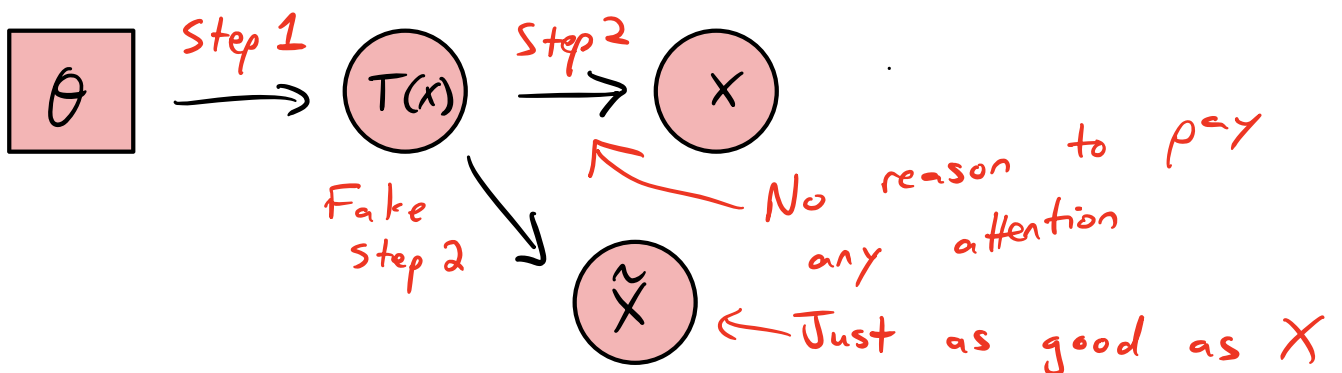
Sufficiency Principle

If $T(X)$ is sufficient for $\mathcal{P}$ then any statistical procedure should depend on X only through $T(X)$

In fact, we could throw away X and generate a new $\tilde{X} \sim P(X|T)$ and it would be just as good as X since $\tilde{X} \sim P_\theta$

$\leftarrow$ no θ

In graphical model form:



Order Statistics

For $x_1, \dots, x_n \in \mathbb{R}$, define order statistics

$$\min_i x_i = x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)} = \max_i x_i$$

Ex (iid sampling on $\mathbb{R}$) $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} P_\theta$,

any model $\mathcal{P} = \{P_\theta^n : \theta \in \Theta\}$ on $\mathcal{X} \subseteq \mathbb{R}$

P_θ^n invariant to perm.s of $X = (X_1, \dots, X_n)$

$\Rightarrow$ All permutations of x are equally likely

$\Rightarrow$ Order statistics $S(X) = (X_{(i)})_{i=1}^n$ sufficient

$X \rightsquigarrow S(X)$ forgets orig. ordering of observations

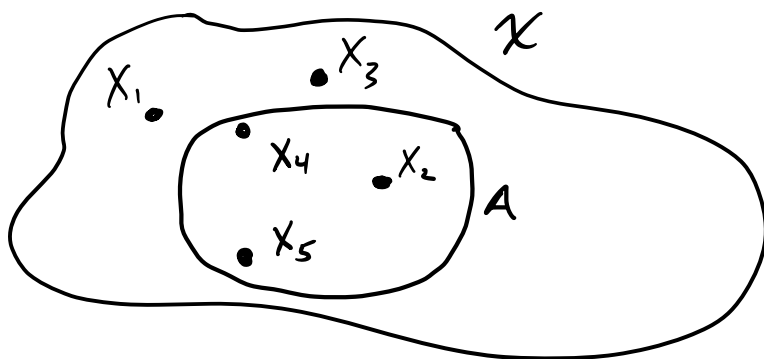
Empirical Distribution

Order statistics depend on total ordering of X
What about more general sample space?

Define Dirac measure $\delta_x(A) = 1_{\{x \in A\}}$

Empirical distribution $\hat{P}_n(\cdot) = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}(\cdot)$

random measure on X , determined by sample



$$\hat{P}_n(A) = \frac{3}{5}$$

Ex (iid sampling) $X_1, \dots, X_n \stackrel{iid}{\sim} P_\theta$

any model $\mathcal{P} = \{P_\theta^n : \theta \in \Theta\}$ on any X

$\hat{P}_n$ is sufficient

$X \rightsquigarrow \hat{P}_n$ records which values observed,
how many times

Minimal Sufficiency

Consider $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\theta, 1)$

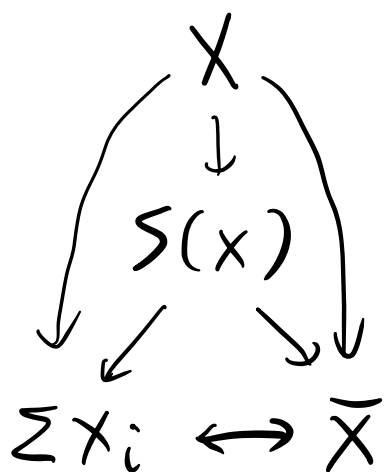
$$T(X) = \sum X_i \quad \text{sufficient}$$

$$\bar{X} = \frac{1}{n} \sum X_i \quad \text{also}$$

$$S(X) = (X_{(1)}, \dots, X_{(n)}) \quad \text{too}$$

$$X = (X_1, \dots, X_n) \quad \text{too}$$

Which can be recovered from which others?



these can be compressed further

These are the most compressed. Are they as compressed as possible?

Prop If $T(X)$ is sufficient and $T(X) = f(S(X))$
then $S(X)$ is sufficient

Proof : $p_{\theta}(x) = g_{\theta}(T(x)) h(x)$
 $= (g_{\theta} \circ f)(S(x)) h(x) \quad \square$

Definition: $T(X)$ is minimal sufficient if

- 1) $T(X)$ is sufficient
- 2) For any other sufficient $S(X)$,
 $T(X) = f(S(X))$ for some f
(a.s. in $\mathcal{P}$)

So, no matter how many more suff. stats we add
to our diagram, they will all have arrows
pointing to ΣX_i

Bayesian Interp.

Bayesian inference: model $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$

prior $\theta \sim \pi$ dist. on Θ

data $X|\theta \sim p_\theta$

$$\leadsto \text{posterior} \quad \pi(\theta|x) = \frac{\pi(\theta) p_\theta(x)}{\int \pi(\xi) p_\xi(x) d\xi}$$

If $T(x)$ sufficient:

$$\pi(\theta|x) = \frac{\pi(\theta) g_\theta(T(x)) \cancel{h(x)}}{\int \pi(\xi) g_\xi(T(x)) \cancel{h(x)} d\xi}$$

$\leadsto$ Posterior only depends on $T(x)$

When do we have $\pi(\theta|x) \equiv \pi(\theta|y)$ (= as fns of θ)

Note: $\pi(\theta|x) \propto_\theta \pi(\theta) p_\theta(x)$

$\leadsto \pi(\theta|x) \equiv \pi(\theta|y)$ iff $\frac{p_\theta(x)}{p_\theta(y)} = \text{const. in } \theta$ (on $\text{supp}(\pi)$)

IFF $\frac{p_\theta(x)}{p_\theta(y)} = \text{const in } \theta$ (same "likelihood shape")

- x and y $\leadsto$ same posterior regardless of π
- x and y are equivalent as evidence ($x \equiv_{\mathcal{P}} y$)

Recognizing minimal sufficiency

Assume $\mathcal{P}$ has densities p_θ , sample space $\mathcal{X}$

Define equivalence relation on $\mathcal{X}$:

$$x \equiv_{\mathcal{P}} y \text{ if } \frac{p_\theta(x)}{p_\theta(y)} \text{ doesn't depend on } \theta$$

Note any sufficient statistic T can only collapse together equivalent values: if $T(x) = T(y) = t$

$$\frac{p_\theta(x)}{p_\theta(y)} = \frac{\mathbb{P}_\theta(X=x, T(X)=t)}{\mathbb{P}_\theta(X=y, T(X)=t)} = \frac{\mathbb{P}(X=x \mid T(X)=t)}{\mathbb{P}(X=y \mid T(X)=t)}$$

So, for any sufficient stat $T(x)$, $T(x) = T(y) \Rightarrow x \equiv_{\mathcal{P}} y$

For minimal sufficient stats, the reverse implication also holds:

Theorem (Bahadur) $T(X)$ is minimal sufficient if

$$X \equiv_p Y \Leftrightarrow T(X) = T(Y)$$

Interp: a minimal sufficient stat. collapses the sample space into exactly these equiv. classes.

Proof: First show $T(X)$ sufficient:

For any x with $T(x) = t$, we have

$$P_{\theta}(X = x \mid T(X) = t) = \frac{p_{\theta}(x)}{\sum_{z: T(z) = t} p_{\theta}(z)} = \frac{1}{\sum_{z: T(z) = t} p_{\theta}(z) / p_{\theta}(x)}$$

which doesn't depend on θ because

$$T(z) = t = T(x) \Rightarrow p_{\theta}(z) / p_{\theta}(x) \text{ doesn't depend on } \theta$$

Next assume $S(X)$ sufficient. If $S(x) = S(y) = s$

then $x \equiv_p y$ so $T(x) = T(y)$. Set $f(s) = T(x)$.

Any other z with $S(z) = s$ has

$$z \equiv_p x \Rightarrow T(z) = T(x) = f(s) = f(S(z)).$$

(log-) Likelihood functions

Definition

Assume $\mathcal{P} = \{P_\theta : \theta \in \Theta\}$ has densities $p_\theta(x)$

The likelihood function is the (random) function

$$\text{Lik}(\theta; X) = p_\theta(x)$$

function of θ data x determines which function function of x with parameter θ

The log-likelihood function is its log:

$$l(\theta; x) = \log \text{Lik}(\theta; x)$$

Note if $x \equiv_P y$ is same as saying

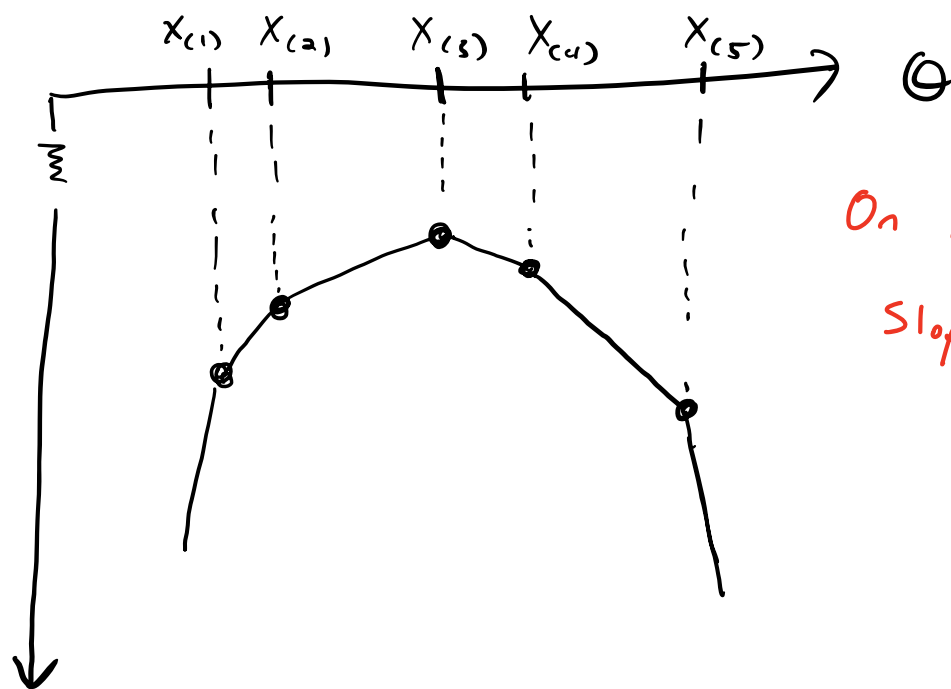
$$l(\theta; x) - l(\theta; y) = \frac{p_\theta(x)}{p_\theta(y)} \text{ is } \underline{\text{constant}}$$

Ex Laplace location family

$$X_1, \dots, X_n \stackrel{iid}{\sim} p_{\theta}^{(1)}(x) = \frac{1}{2} e^{-|x-\theta|}$$

$$l(\theta; x) = - \sum_{i=1}^n |x_i - \theta| - n \log 2$$

Piecewise linear in θ , knots at $x_{(i)}$



On $[x_{(k)}, x_{(k+1)}]$,

$$\text{Slope} = n - 2k$$

$$l(\theta; x) = l(\theta; y) + \text{const} \Leftrightarrow X, Y \text{ same order statistics}$$

$\Rightarrow$ order stats are minimal suff.